

Fractions with Different Denominators: Comparison, Addition and Subtraction

Unit 3 · Fraction operation · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A, Unit 3 + Modern Education 5A, Unit 10

Core Traps: □ T9 Fraction Operations - LCM Error Finding · The numerator forgets to expand simultaneously after the common division · The result is not reduced

SSPA Association: ● **High frequency** The score test is required every year, accounting for about 15-20% of Paper 1

Prerequisite knowledge: P4 Addition and subtraction of fractions with the same denominator · LCM calculation
• Equivalent fractions and extended fractions

Course goal: ❶ Comparison of fractions with different denominators (common fraction → ratio) ❷ Addition of different denominators (three-step method for common denominators) ❸ Subtraction of different denominators ❹ Mixed application of addition and subtraction of fractions

Next class connection: Class 8 (In-depth practice of adding fractions with different denominators) - This class lays a solid foundation for comparison and basic addition and subtraction

Student Name:

Class:

Date:

Time Spent:

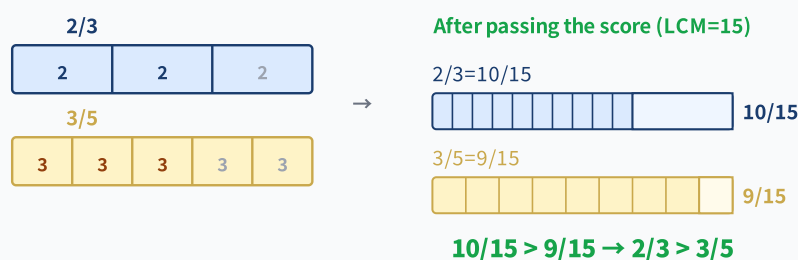
I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate LCM(4, 6) = ?	Basic	
2	Calculate LCM(3, 5) = ?	Basic	
3	Compare $\frac{2}{3}$ and $\frac{3}{5}$ which one is bigger? Write down the judgment process.	Advanced	
4	Calculate $\frac{3}{7} + \frac{2}{7} = ?$ (Adding with the same denominator, P4 basics)	Basic	
5	Calculate $\frac{5}{9} - \frac{2}{9} = ?$ (Subtraction with the same denominator, P4 basic)	Basic	

II.Core Knowledge + Worked Examples

Knowledge point 1: Comparison of fractions with different denominators - the ratio of the common denominator ● SSPA

- ① Different denominators = different sizes of each part, the numerators cannot be directly compared
- ② **Three-step comparison method:** Find LCM → common denominator (expand the denominator to the same denominator) → compare the sizes of the molecules
- ③ After the common denominator, the fraction with a larger molecule is larger; the fraction with a smaller molecule is smaller
- ④ Comparison of three or more fractions: all common denominators to the same denominator (LCM of all denominators), and then arrange the molecules
- ⑤ **Trap T9:** LCM error finding → The basics of general division are wrong → The whole question is wrong!



Example 1

Comparing $\frac{3}{4}$ and $\frac{5}{6}$, which one is larger? Write down the general process.

Example 2 (Ordering three fractions)

Arrange $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$ from largest to smallest.

⚠ The most frequent error: compare the numerators directly without common denominators! Different denominators → different sizes of each portion → the numerators cannot be directly compared!

Knowledge Point 2: Addition of Fractions with Different Denominators - Three-Step Method for Common Fractions ● SSPA Required Exam

- ① **Why can't you add it directly?** The denominator represents the serving size. Different denominators = different sizes of parts, you cannot add the numerator directly
- ② **Three steps:** Find LCM (least common multiple of the denominator) → Expand the fraction (the numerator is expanded simultaneously) → The denominator remains unchanged and the numerators are added
- ③ The answer must be **Reduction** (the simplest fraction); improper fraction must be **Convert mixed fractions**
- ④ **Trap T9:** When converting fractions, only change the denominator and forget to expand the numerator simultaneously!

①

Find LCM

Find the lowest common multiple using short division

②

General/expanded

The numerator and denominator change simultaneously to LCM

③

calculate

Adding numerators with the denominator unchanged

④

reduce

Simplest false fraction → mixed fraction

□ Example of trap detonation (the most important demonstration in this class)

calculate: $\frac{1}{2} + \frac{1}{3} = ?$

✗ Common mistakes (80% students)

$$\frac{2}{5}$$

Direct numerator plus numerator, denominator plus denominator

✓ Correct solution

$$\frac{5}{6}$$

$$\text{LCM}(2,3)=6 \rightarrow \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

Example 3

Calculation: $\frac{2}{3} + \frac{3}{4} = ?$ (Write down the four steps of ①LCM ②general split ③calculation ④reduction)

🧠 **Tip: "First understand the LCM, then multiply the numerator and keep the denominator unchanged. The answer must be simplified."**

⚠ The second most frequent error: only changing the denominator when dividing the whole number, forgetting to expand the numerator simultaneously. "The molecules follow and ride"!

⚠ The third most frequent error: the answer is not reduced. For example, 612 must be written as 12! One point will be deducted if the test is not simplified.

Knowledge point 3: Subtraction of fractions with different denominators - exactly the same steps as addition ● SSPA required test

- ① Subtraction and addition use **exactly the same general division steps**- the only difference: the last step is "numerator subtraction" instead of "numerator addition"
- ② **Notes on subtraction:** Make sure that the numerator of the minuend $\geq$ the numerator of the minuend (after common division). If it is not enough $\rightarrow$ Borrowing is required (with fractions)
- ③ The answer also requires **Reduction** to the simplest fraction

❏ Trap example (subtraction)

calculate: $\frac{5}{6} - \frac{1}{4} = ?$

✗ Common mistakes

$$\frac{4}{2} = 2$$

Direct numerator minus numerator, denominator minus denominator

✓ Correct solution

$$\frac{7}{12}$$

LCM(6,4)=12 $\rightarrow \frac{10}{12} - \frac{3}{12} = \frac{7}{12}$

Example 4

calculate: $\frac{7}{8} - \frac{2}{3} = ?$

Knowledge point 4: Mixed addition and subtraction of fractions + application questions ● SSPA must take the exam

- ① **Mixed addition and subtraction:** First divide all to the same denominator (LCM of all denominators), then calculate the numerator by +, -, and finally reduce
- ② **Four steps to solve the application problem:** Circle the key words $\rightarrow$ Column formula (use + or - for judgment) $\rightarrow$ Calculate the common denominator $\rightarrow$ Write a complete answer
- ③ Keywords: "Total" "Total" = Addition; "remaining" and "difference" = subtraction; "more than..." $\rightarrow$ Find the number first and then add

Example 5

Calculation: $\frac{3}{4} + \frac{1}{6} - \frac{1}{3} = ?$ (mixing of addition and subtraction of three fractions)

Example 6 (example of application questions)

A bottle of juice contains $\frac{3}{4}$ liters. Xiao Ming drank $\frac{1}{3}$ liters, and his mother poured in $\frac{1}{2}$ liters.
How many liters of juice are there in the bottle now? (Answer with mixed fractions)

III. Lesson Layered Synchronization Practice

Basic layer (total 6 questions, everyone must do)

#	Question	Difficulty	Working Space
7	Comparing $\frac{1}{2}$ and $\frac{2}{5}$, which one is larger? (Write down the process of passing grades)		
8	calculate: $\frac{1}{3} + \frac{1}{6} = ?$		
9	calculate: $\frac{1}{4} + \frac{1}{2} = ?$		
10	calculate: $\frac{3}{5} - \frac{1}{10} = ?$		
11	calculate: $\frac{2}{3} - \frac{1}{6} = ?$		
12	Comparing $\frac{3}{8}$ and $\frac{5}{12}$, which one is larger? (LCM = 24)		

Advanced layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
13	calculate: $\frac{2}{5} + \frac{3}{10} = ?$		
14	calculate: $\frac{3}{4} - \frac{1}{6} = ?$		
15	calculate: $\frac{5}{6} - \frac{1}{3} = ?$		
16	Comparing $\frac{4}{7}$ and $\frac{5}{9}$, which one is larger? (Pass score required, LCM = 63)		
17	Calculation: $\frac{1}{2} + \frac{1}{3} - \frac{1}{6} = ?$ (mix of addition and subtraction)		

 challenge layer (total 5 questions,   choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
18	Sorting the three scores: Arrange $\frac{5}{6}$ 、 $\frac{3}{4}$ 、 $\frac{7}{12}$ from large to small.		
19	Calculation: $\frac{2}{3} + \frac{1}{4} + \frac{1}{6} = ?$ (Adding three fractions, LCM = 12)		
20	Calculation: $\frac{7}{8} - \frac{1}{6} - \frac{1}{4} = ?$ (Subtraction of three fractions)		
21	Xiao Ming originally had $\frac{3}{4}$ cakes, ate $\frac{1}{3}$ cakes, and bought $\frac{1}{2}$ cakes. How many cakes are there now?		
22	Among the three fractions SEP , what is the difference between the largest and the smallest? $\frac{2}{3}$ 、 $\frac{5}{12}$ 、 $\frac{1}{2}$, what is the difference between the maximum and minimum?		

IV.applicationquestionpractice (text questionspecial topic, SSPAMust-Test)

Application problem solving strategies SSPA

- ① **Circle keywords:**"total" "total" → addition; "remaining" "difference" "less than" → subtraction
- ② **Column formula:**Use fraction symbols to write the complete formula
- ③ **Common fraction calculation:**Find LCM → Expand → Add and subtract molecules → Reduction
- ④ **Write an answer sentence:**"Answer:... (Unit)" - Step points will be deducted for not writing an answer sentence!

applicationquestion (total 6 questions, all must do, column → find a common denominator → calculate → answer sentence)

#	Question	Difficulty	Working Space
23	Xiao Mei drank $\frac{1}{4}$ liters of orange juice, and Xiao Ming drank $\frac{1}{2}$ liters. How many liters did they drink in total?		
24	A box of chocolate, my sister ate $\frac{2}{5}$ boxes, my brother ate $\frac{1}{3}$ boxes. How many boxes did the two of them eat in total?		
25	A rope is $\frac{5}{6}$ meters long, cut off $\frac{1}{4}$ meters. How many meters are left?		
26	The original water bottle contains $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters into it. How many liters are there now? (Answer with mixed fractions)		

27	The cake shop sold $\frac{2}{5}$ cakes in the morning and $\frac{1}{3}$ cakes in the afternoon. How many were sold in total throughout the day?		
28	A bottle of juice contains $\frac{7}{8}$ liters. Xiao Ming drank $\frac{1}{4}$ liters and Xiaohua drank $\frac{1}{3}$ liters. How many liters are left? (subtract twice)		

V. Ultimate challenge area (total 3 questions, choose do, SSPA finale level)

#	Question	Difficulty	Working Space
 1	Calculation: $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} = ?$ (Adding four fractions with different denominators - find LCM(2,3,4,6))		
 2	Compare the equation $A = \frac{3}{4} + \frac{1}{6}$ with the equation $B = SEP $. Which result is greater? How much difference? $\frac{2}{3} + \frac{1}{4}$. Which result is greater? How much difference?		
 3	A rope is 2 meters long. $\frac{1}{3}$ was used on the first day, and the remaining $\frac{1}{2}$ was used on the second day. How many meters are left? (Hint: Remaining after the first day = $2 - \frac{2}{3} = \frac{4}{3}$ meters; used the remaining on the second day $\frac{1}{2} = \frac{4}{3} \times \frac{1}{2}$)		

VI. Class after homework

Basic must-do questions (total 5 questions, must write find a common denominator step)

#	Question	Difficulty	Working Space
H1	Comparing $\frac{2}{3}$ and $\frac{3}{4}$, which one is larger? Write down the general process.		
H2	calculate: $\frac{1}{2} + \frac{1}{6} = ?$		
H3	calculate: $\frac{1}{3} + \frac{1}{9} = ?$		
H4	calculate: $\frac{3}{4} - \frac{1}{2} = ?$		
H5	calculate: $\frac{5}{6} - \frac{1}{3} = ?$		

Advanced choose do question (total 3 questions, choose do)

#	Question	Difficulty	Working Space
H6	calculate: $\frac{2}{3} + \frac{1}{4} = ?$		
H7	One cake, Xiao Ming ate $\frac{1}{4}$ pieces, and Xiaohua ate $\frac{1}{3}$ pieces. How many cakes did the two of them eat together?		

H8	There are $\frac{5}{6}$ liters in a bucket of water, and $\frac{1}{4}$ liters are used. How many liters are left?		
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VII. The Lesson core common errors summary

✅ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Direct addition and subtraction with different denominators : $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$	Different denominators → Be sure to connect the denominators first! Find LCM → Expand → Calculate again
2	After splitting, the molecules forget to expand simultaneously.	"Multiply the numerator" - multiply the denominator by the same number and multiply the numerator by the same number
3	The answer is not reduced : $\frac{6}{12}$ when the answer	The last step must be to check the common factors and convert them into the simplest fraction
4	Improper fractions are not converted to mixed fractions	Numerator > Denominator → Convert into mixed numbers (subject to the requirements of the division test)
5	LCM error finding (replace LCM with larger common multiple)	Short division confirmed to be the true "least" common multiple
6	When comparing, only look at the numerator and not the denominator	Different denominators → Different sizes of parts → Comparison must be made only after dividing the parts
7	Missing sentences/units in application questions	You must write "Answer:..." + unit (liter, meter, unit), otherwise step points will be deducted